



BETTA FISH · BETTA SPLENDENS

# Keep your betta the right way from day one.

A complete, fact-checked printable manual: tank size and heat, the nitrogen cycle that decides everything, feeding without overfeeding, health triage, and the routines that keep a betta thriving for its full lifespan.

## INSIDE

- ✓ Every temperature in °F and °C, and every parameter with a target number
- ✓ A full fishless cycling walkthrough and a testing schedule
- ✓ Five health pages, from fin rot to velvet, with dosing and duration limits
- ✓ Printable emergency card, pet-sitter sheet, and blackout plan

37 pages

Beginner

Print or tablet

Version 2.2 · Sep 2026



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Every page is written to be used, not just read once.

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## START HERE

# How to use this package

Three ways to get value from it, depending on where you are right now.

## If you haven't set up yet

Read pages 4 through 12 in order before buying anything, and before buying the fish. The single most important thing in this book is that the tank is cycled before a betta goes in it, and cycling takes weeks, not an afternoon. Almost every betta that dies in its first month dies of something that started here. Use the Budget & Shopping List on page 25 to buy once, then start the cycle on page 7 while the fish is still in the shop.

## If your betta is already set up

Go straight to the Setup Checklist and Water Targets on page 23 and audit what you have. If the tank has no heater, no filter, or has never been tested with a liquid test kit, fix that this week rather than reading further. If your betta already looks unwell, start at Health Red Flags on page 16, but test the water first: the answer is in the test tube far more often than it is in the medication cabinet.

## Print these first

- |                                                                    |                                                                     |
|--------------------------------------------------------------------|---------------------------------------------------------------------|
| <input type="checkbox"/> Page 23: Setup Checklist & Water Targets  | <input type="checkbox"/> Page 31: Pet-Sitter Sheet                  |
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### WHY THIS PACKAGE EXISTS

The betta is sold worse than almost any other pet: in a cup, next to an unheated bowl, described as a fish that wants a small space. None of that survives contact with the research, and the peer-reviewed enrichment work on page 14 says the opposite outright. This package reflects current tank-size, filtration, and enrichment standards, cross-checked against veterinary and aquarium-science sources listed on page 36, not recycled pet-aisle copy.

### NOT A SUBSTITUTE FOR

An aquatic veterinarian. Fish vets are rarer than reptile or avian vets, and most betta problems really are solved by correcting water quality rather than by treatment, but "rare" is not "nonexistent." Find out now whether one practices within reach of you, and write the number on page 24 before you need it.

## SECTION 01

# Quick profile

Everything at a glance before you read the detail.

|                     |                                                                                |
|---------------------|--------------------------------------------------------------------------------|
| Scientific name     | <i>Betta splendens</i> , also called the Siamese fighting fish                 |
| Adult size          | 2.5 to 3 in (6 to 7.5 cm) body. Male finnage is fin, not extra size            |
| Captive lifespan    | 2 to 4 years, up to 5 with excellent water quality. Often months old when sold |
| Native range        | Shallow rice paddies, floodplains and slow streams in Southeast Asia           |
| Diet type           | Obligate carnivore, a surface ambush feeder. Pellet staple, never flakes       |
| Tank minimum        | 5 gal (19 L) heated, filtered, cycled and lidded. 10 gal (38 L) is far easier  |
| Temperature         | 78 to 80°F (25.5 to 26.5°C), held steady. Never room temperature (page 5)      |
| The one hard rule   | Ammonia and nitrite read 0 ppm, always. Anything else is an emergency (page 8) |
| Housing             | One male per tank. Surface access is a physiological need (page 5)             |
| Handling difficulty | Beginner, but observation only. Never bare hands (page 15)                     |
| Conservation status | Vulnerable (IUCN) in the wild, despite huge numbers in captivity               |

## Cost overview

### Initial setup

\$157 to \$336 for a complete cycled setup including the test kit, plus \$5 to \$30 for the fish itself and up to about \$55 for a rare variety.

### Monthly ongoing

\$10 to \$25 for food, conditioner, test reagents, electricity, and a medication reserve. Roughly \$120 to \$300 a year.

#### WHERE THE MONEY ACTUALLY GOES

Not the fish, and not the food. A betta eats a few pellets a day, so the recurring cost is a heater running around the clock and fresh test reagents. The upfront figure looks high next to a \$6 fish because it includes the two things the pet-aisle bundle leaves out, a heater and a real filter, and those decide whether the fish lives two months or four years.

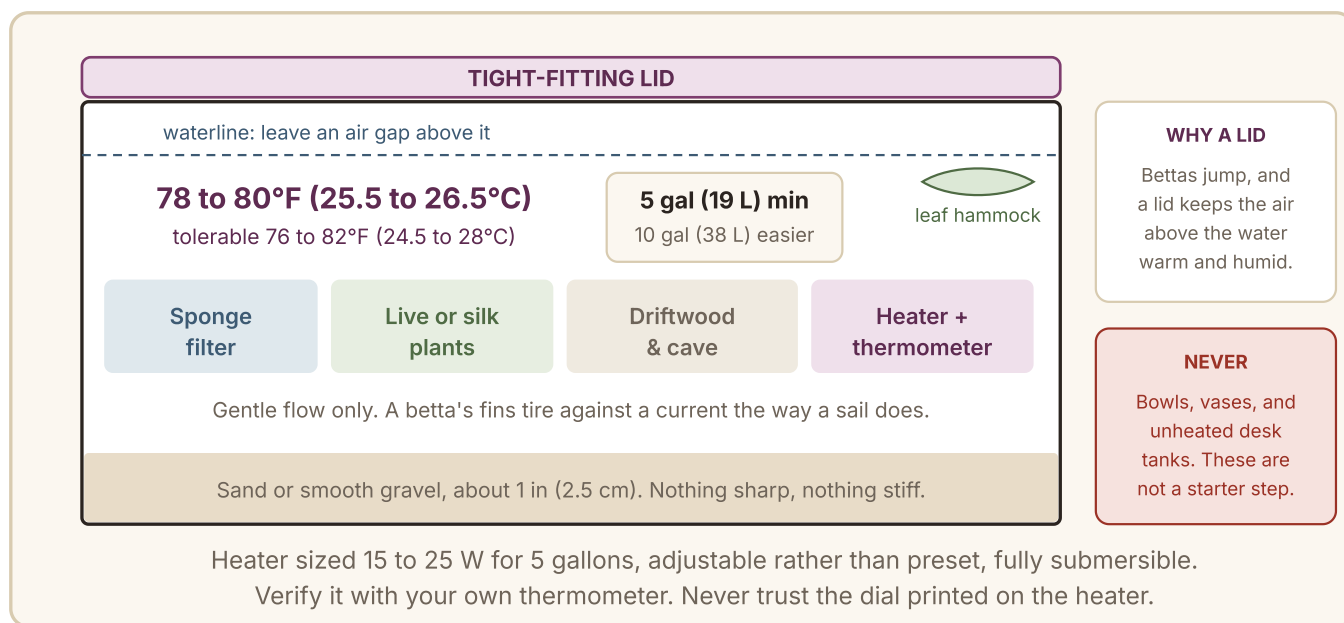
## Fun facts worth knowing

- They breathe air. The labyrinth organ above the gills takes oxygen straight from the surface, an adaptation to warm paddy water holding very little dissolved oxygen. It also means a betta blocked from the surface can drown.
- Males build nests out of spit. A bubble nest is a raft of saliva-coated bubbles built at the surface to hold eggs. Captive males build them with no female present, and it reads as a comfortable, well-warmed fish.
- The aggression is partly built, not born. Research found that the timing of isolation from an enriched environment shapes later aggression, so your fish's temperament was set before you bought it.

## Tank, heater & lid

Five gallons (19 L) is the current minimum, and it is a floor rather than a target. Ten gallons (38 L) is the size most keepers wish they had started with: more water dilutes waste further, holds temperature more steadily, and gives you hours of margin when something goes wrong instead of minutes. A standard 5 gallon measures about 16×8×10 in (40×20×25 cm), a 10 gallon about 20×10×12 in (50×25×30 cm). Length and surface area matter more than depth for a fish that has to reach the top to breathe.

A betta cannot be kept at room temperature. It is a tropical fish from water that sits in the high seventies year round, and an unheated tank in a heated house still drifts several degrees every night. Cold water suppresses immune function and digestion, which is why so many "mystery" declines are really slow chilling.



### NEVER

Run a betta in an unheated container, or trust a preset "tropical" heater without checking it against a separate thermometer. A stuck-on heater cooks a 5 gallon tank in hours, and a stuck-off one chills it overnight. The thermometer is what tells you which is happening.

# Filtration & the nitrogen cycle

The two most important pages in this package. Everything on them happens before the fish comes home.

## The filter

A sponge filter driven by a small air pump is the standard for this species, for one specific reason: a betta's fins catch water like a sail, and the outflow a tetra ignores will exhaust a betta and fray its fins over weeks. A sponge filter provides biological filtration with almost no current. If you use a hang-on-back filter instead, baffle the outflow with a piece of filter foam or a cut plastic bottle until the surface is barely rippling.

Fit a check valve on the airline between the pump and the tank. Without one, a power cut lets water siphon back down the tube and out onto the floor, which is how a small outage becomes a flood and an empty tank.

## What the filter actually does

Almost nothing you can see. The sponge is a surface for bacteria to live on, and those bacteria are the filter. They convert the ammonia in fish waste and uneaten food into nitrite, then into far less harmful nitrate, which you remove with water changes. That colony is what "cycled" means, and it is why a tank that looks spotless can still be lethal.

| STAGE      | WHAT IS HAPPENING                                                                                              | WHAT YOU SEE ON A TEST                   |
|------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1. Ammonia | Waste and leftover food break down into ammonia, which burns gills at any level                                | Ammonia climbs, nitrite and nitrate at 0 |
| 2. Nitrite | <i>Nitrosomonas</i> bacteria establish and convert ammonia to nitrite, which blocks oxygen uptake in the blood | Ammonia falls, nitrite spikes            |
| 3. Nitrate | A second group of bacteria converts nitrite to nitrate, which is mildly toxic and easy to dilute               | Nitrite falls to 0, nitrate appears      |
| 4. Cycled  | Both colonies keep pace with the tank's daily waste load                                                       | Ammonia 0, nitrite 0, nitrate present    |

### CYCLE WITH THE PLANTS ALREADY IN

Put the plants, driftwood, and decor in before you start. Plants take up ammonia directly, the decor gains its own share of the colony, and you avoid disturbing a settled tank later. It costs nothing to do it in this order.

## Fishless cycling, step by step

Weeks of work with no fish in the tank. This is the part that gets skipped, and it is the part that decides everything.

1. Set the tank up completely: substrate, plants, decor, heater at 78 to 80°F (25.5 to 26.5°C), filter running. Warm water cycles faster, so the heater goes in on day one.
2. Fill with tap water dosed with dechlorinator. Chlorine and chloramine kill the bacteria you are trying to grow.
3. Add an ammonia source: pure household ammonia, no surfactants or perfume, dosed to 2 to 4 ppm. Test to confirm rather than counting drops.
4. Add bottled nitrifying bacteria, or better, a squeeze of used sponge or gravel from an established healthy tank. Seeded media cuts weeks off this.
5. Test ammonia and nitrite every 2 to 3 days. Re-dose ammonia back to 2 ppm whenever it falls to 0, so the growing colony keeps being fed.
6. The tank is cycled when it can take a 2 ppm ammonia dose and read **0 ammonia and 0 nitrite 24 hours later, with nitrate present**. Not "close to zero." Zero.
7. Change enough water to bring nitrate under 20 ppm, temperature-matched, and only then bring the fish home.

Expect 3 to 6 weeks with bottled bacteria, longer without. There is no way to rush the biology, only ways to seed it.

### IF THE FISH IS ALREADY HOME AND THE TANK ISN'T CYCLED

This is a fish-in cycle and it is a rescue, not a plan. Test ammonia and nitrite daily. Watch from 0.1 ppm and change water at 0.25 ppm, scaling the change to the reading: 25 to 30% just over the line, 50% or more once either number is clearly climbing. Use a conditioner that detoxifies ammonia between changes, feed lightly every other day, and add seeded media if you can get it. Expect this to run several weeks and to need attention every single day.

## When the cycle stalls

| WHAT YOU SEE                                   | WHAT IS USUALLY HAPPENING                                                                                                                                    |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ammonia sits high for weeks, no nitrite at all | Chlorine or chloramine still present, an ammonia product with additives, or the tank is too cold. Re-dose dechlorinator, confirm 78 to 80°F (25.5 to 26.5°C) |
| Nitrite spikes and then stays there            | Normal, and the longest stage. High ammonia holds it back, so stop re-dosing above 2 ppm until nitrite falls                                                 |
| Everything reads 0, including nitrate          | Usually nothing has started yet. Confirm the ammonia dose actually registered on a test before assuming the tank is cycled                                   |
| pH crashes and the cycle stops dead            | The bacteria consume carbonate hardness. Very soft water runs out of buffer, pH falls, and the colony stalls. A partial water change usually restarts it     |
| It cycled, then broke after a filter clean     | The colony lives on the sponge. Rinsing it under the tap kills it. Rinse in old tank water only, and only when flow drops                                    |

## Water quality & testing

Most betta illness is a water problem wearing a costume. These are the numbers behind that.

| PARAMETER   | TARGET                      | WHY IT MATTERS                                                                                                                  |
|-------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Temperature | 78 to 80°F (25.5 to 26.5°C) | Cold water suppresses immune function and digestion, and slows recovery from everything else on this list                       |
| pH          | 6.5 to 7.5                  | Wide tolerance. Stability matters far more than the exact figure                                                                |
| Ammonia     | 0 ppm                       | Even 0.25 ppm causes gill damage. Any detectable reading is an emergency                                                        |
| Nitrite     | 0 ppm                       | Blocks oxygen uptake in the blood. Toxic at any measurable level                                                                |
| Nitrate     | Under 20 ppm                | The normal end product of a healthy cycle, controlled by water changes                                                          |
| GH / KH     | Moderate, stable            | Secondary. Very soft, poorly buffered water lets pH crash after a change, which is worse than a steady pH slightly out of range |

### ZERO MEANS ZERO

Ammonia and nitrite are the only two numbers here with no safe nonzero baseline. Temperature and pH have ranges. These do not. A cycled, established tank reads exactly 0 ppm for both, consistently, and any other reading is something to act on today rather than watch.

## Test with reagents, not strips

A liquid reagent kit is the only reliable way to know whether a tank is cycled. Paper strips are notoriously imprecise, especially at the low end of the ammonia and nitrite scales, which is exactly the range where the answer matters. One kit lasts a long time and works out cheaper per test than strips do.

| WHEN                        | WHAT TO TEST                                                                                                          |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| New or recently set up tank | Ammonia and nitrite every 1 to 2 days until both read 0 for a full week                                               |
| Established, cycled tank    | Ammonia, nitrite, and nitrate weekly, on the same day each week                                                       |
| Anything looks off          | Test immediately. Clamped fins, lethargy, and appetite loss usually show up in the water before they show on the fish |
| After a change over 30%     | A quick recheck confirms the new water didn't bring chlorine or chloramine in with it                                 |
| After adding anything       | New decor, a new plant, a medication, or a filter change: test for the next few days                                  |

## Water changes & keeping the cycle

The single most useful habit in fishkeeping, and the easiest one to do in a way that undoes itself.

### The weekly water change

1. Change 20 to 30% of the volume weekly. In a 5 gallon that is roughly 1 to 1.5 gallons (4 to 6 L).
2. Vacuum the substrate as you siphon, working over a different section each week rather than the whole floor at once.
3. Prepare the replacement water in a dedicated bucket, dose the dechlorinator into it, and match the temperature to the tank by feel and then by thermometer. A few degrees of difference is a real stressor at this size.
4. Pour it in slowly against the glass or onto a plate rather than straight onto the substrate or the fish.
5. Rinse the filter sponge in the bucket of old tank water, gently, and only when flow visibly drops. Never under the tap.

### When to change more than 30%

| SITUATION                                        | CHANGE                                                                                |
|--------------------------------------------------|---------------------------------------------------------------------------------------|
| Ammonia or nitrite above 0 ppm                   | 25 to 50% now, and again the next day if it has not come back to 0                    |
| Nitrate over 20 ppm                              | Change more, or change more often, rather than doing one very large change            |
| Overfeeding accident, food left on the substrate | Siphon the food out first, then change 25%                                            |
| Medicating, at the end of a course               | Follow the label. Usually a large change plus fresh carbon once treatment is finished |
| Routine weekly maintenance                       | 20 to 30%. Bigger is not better here                                                  |

#### TOPPING UP IS NOT A WATER CHANGE

Water evaporates, dissolved waste does not. Adding fresh water to replace what evaporated raises the level and dilutes nothing, so nitrate keeps climbing while the tank looks full. Top up when the level drops, and change water on schedule regardless. Keep the lid on and there will be far less of both to do.

#### WHAT RESETS A CYCLE

A 100% water change, scrubbing or replacing the filter sponge, rinsing media in chlorinated tap water, or deep-cleaning the whole tank at once. Each of these throws out the bacteria colony and puts you back at day one of page 7, with a fish already in the tank. Buckets, siphons, and sponges used for the aquarium stay aquarium-only, and never meet soap.

# Diet & feeding schedule

Overfeeding is the most common mistake in betta keeping, and it is the one that fouls the water too.

Bettas are obligate carnivores and surface ambush feeders: in the wild they take insects, larvae, and small crustaceans off the top of the water. In a tank the staple is a betta-specific high-protein pellet, not tropical flakes and not goldfish food, both of which have the wrong nutrient profile and far more filler. Look for a pellet that lists brine shrimp, krill, or fish meal near the top of the ingredients and runs roughly 30 to 35% protein and around 5% fat.

## Portion, which matters more than frequency

A betta's stomach is roughly the size of its own eye. That is the actual basis for the standard guidance of 2 to 4 pellets per meal, and it is why a fish that looks hungry is usually just a fish that always looks hungry. Depending on pellet size and brand this can run anywhere from 2 to 3 up to 6 to 7, so the more reliable rule is whatever gets cleared within 1 to 2 minutes. Remove anything still floating after that.

### THE ROUTINE

Once or twice a day, at least 8 hours apart if you feed twice, with one full fasting day a week. Pick a schedule and hold it. Consistency does more here than the exact number of meals.

### PRE-SOAK EXPANDING PELLETS

If a pellet swells in water, it swells in the fish too. Soak it in a spoonful of tank water for a minute first. This alone prevents a large share of bloat and constipation cases.

### MOVE THE FEEDING SPOT

Drop food in a different part of the tank each day and rotate what you offer. It is the cheapest enrichment there is, and the fish general enrichment literature backs feeding variation directly.

## By life stage

| STAGE                | FEEDING                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| Fry, freshly hatched | Infusoria several times a day, before they are free-swimming                                              |
| Fry, free-swimming   | 4 to 5 small meals daily                                                                                  |
| Weeks 3 to 4         | 3 to 4 small meals daily                                                                                  |
| Juvenile             | 2 to 3 meals daily, moving onto the adult routine once separated into their own space                     |
| Adult                | Once or twice daily, 2 to 4 pellets per meal, one fasting day a week                                      |
| Overweight adult     | Remove one pellet per meal, hold for 2 weeks, reassess against page 15, and repeat if nothing has changed |

## Why a betta stops eating

Bettas refuse food for six ordinary reasons and one worrying one. Work them in order.

1. **Environmental stress.** A new tank, a recent move, or a disturbance nearby suppresses appetite temporarily.
2. **Poor water quality.** An ammonia, nitrite, or nitrate spike. Check this first, every time, before anything else on the list.
3. **Water too cold.** Below roughly 76°F (24.5°C) metabolism and appetite both slow. Most unheated rooms sit here.
4. **Picky eating.** Bettas reject an unfamiliar brand outright. Try a different food before assuming illness.
5. **Overfeeding, constipation, or swim bladder trouble.** A swollen belly, stringy or absent feces, floating or sinking (page 20).
6. **Illness.** Bacterial infection or parasites, usually alongside clamped fins, white spots, or labored swimming.

### WHEN A FAST BECOMES A SYMPTOM

A healthy adult can physically survive around two weeks without food, but treat that as an emergency ceiling, not a guideline. The point to investigate is **5 to 7 days of refusal** after you have already checked water quality and temperature and tried a different food, and immediately at any point if a physical symptom from page 16 appears alongside it.

## What to try, in order

1. Test ammonia, nitrite, and nitrate, and read the thermometer. Do this before anything else, every time.
2. Change 25% of the water, temperature-matched, whether or not the test showed anything.
3. Look the fish over properly: fins, belly, scales, gills, and how it holds itself in the water.
4. Skip a day. A betta that has been overfed genuinely is not hungry, and a fast day is already part of the schedule.
5. Offer something different: thawed brine shrimp or daphnia will tempt a fish that has gone off pellets.
6. Check what changed around the tank. A new light, a mirror-bright window, a moved position, or a new reflection all suppress appetite.

### TEMPTING A FUSSY BETTA

Feed at the same time each day so hunger lines up with the routine. Pre-soak pellets, which softens them and releases scent. Drop food where the fish already is rather than expecting it to cross the tank. And give a new food three or four tries across a week before deciding it has been rejected: bettas often refuse something unfamiliar twice and then take it.

## Food chart & the never-feed list

Staple every day, occasional a few times a week, rare as a remedy, never at all.

| FOOD                           | TIER       | HOW TO OFFER IT, AND HOW OFTEN                                             |
|--------------------------------|------------|----------------------------------------------------------------------------|
| Betta-specific pellet          | STAPLE     | Backbone of the diet. 2 to 4 pellets a meal, pre-soaked if it expands      |
| Frozen brine shrimp            | OCCASIONAL | 1 to 3 times a week, thawed first. A good option for a fussy eater         |
| Frozen or live daphnia         | OCCASIONAL | 1 to 3 times a week. Mildly laxative, useful during constipation           |
| Mysis shrimp                   | OCCASIONAL | 1 to 3 times a week. Larger than brine shrimp, so offer fewer pieces       |
| Mosquito larvae                | OCCASIONAL | 1 to 3 times a week. As close to the wild diet as anything you can buy     |
| Tubifex worms                  | OCCASIONAL | 1 to 3 times a week, from a reputable source. Rich, so keep portions small |
| Bloodworms                     | LIMIT      | 1 to 2 times a week at most, 1 to 2 worms per meal. Bettas gorge on these  |
| Freeze-dried anything          | LIMIT      | Rehydrate in tank water first. Fed dry it swells inside the fish           |
| Blanched, shelled pea          | REMEDY     | One small piece as a constipation remedy after a 2 to 3 day fast           |
| Tropical flakes, goldfish food | NEVER      | Wrong nutrient profile, too much filler, and they foul the water           |
| Bread and any baked good       | NEVER      | Swells in the gut, almost no nutritional value for a carnivore             |
| Onion and garlic               | NEVER      | Toxic sulfur compounds                                                     |
| Salt as a food, sugary foods   | NEVER      | Aquarium salt is a treatment dosed into water, never a food (page 18)      |
| Processed meats                | NEVER      | Salt, preservatives, and fat a betta cannot process                        |
| Citrus                         | NEVER      | Acidic enough to crash tank pH and burn gills                              |
| Dairy                          | NEVER      | Bettas are effectively lactose intolerant and cannot digest it at all      |

### READING THE LABEL

A good pellet lists a named animal protein first, brine shrimp, krill, or fish meal, not "wheat flour" or "fish derivatives." Target roughly 30 to 35% protein and 5% fat. Size matters too: a pellet the fish struggles to swallow gets spat out and rots on the substrate.

### TWO HABITS WORTH BREAKING EARLY

Feeding because the fish begs at the glass, and topping up food still sitting uneaten. A betta begs indefinitely, and every pellet that reaches the substrate becomes ammonia within a day. Leftovers mean you fed too much.

## Common mistakes

### The mistakes that cost bettas their lives

| MISTAKE                                  | WHAT ACTUALLY HAPPENS                                                                                                                |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Adding the fish to an uncycled tank      | Ammonia has nowhere to go. This is why so many bettas decline in the first two weeks of a brand-new, spotless-looking setup (page 7) |
| No heater                                | Chronic cold suppresses immunity and digestion. The fish does not die of cold, it dies of the infection it could not fight off       |
| A bowl, vase, or the cup it came in      | No filtration, no stable temperature, and restricted space, which the research directly links to hiding and inactivity               |
| Overfeeding                              | Bloating, constipation, swim bladder trouble, and fouled water from what was left uneaten                                            |
| Strong filter flow                       | A betta's finnage catches current. Constant swimming against it exhausts the fish and frays the fins over weeks                      |
| Stiff plastic plants and sharp decor     | Torn fins, which is a real and common injury and an open door for fin rot                                                            |
| Testing with paper strips, or not at all | Imprecise exactly where precision matters, at the low end of ammonia and nitrite                                                     |
| Reaching for medication first            | Most of Section 03 resolves by correcting water and temperature. Medicating a water-quality problem adds a second problem            |
| Using a mirror as a daily toy            | What you are provoking is a threat display. Researchers use exactly this to generate a measurable stress response                    |

### Myths worth retiring

| THE MYTH                                        | WHAT IS ACTUALLY TRUE                                                                                                                                                   |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Bettas live in puddles in the wild"            | They live in shallow, warm, heavily vegetated water that spans large areas. Shallow is not small, and seasonal puddles are where wild bettas die, not where they thrive |
| "They prefer small spaces"                      | Restricted space raised hiding and inactivity in the research. Nothing has ever shown a preference for less room                                                        |
| "They breathe air, so they don't need a filter" | The labyrinth organ solves oxygen. It does nothing about ammonia, which is what the filter is for                                                                       |
| "A betta plant or bowl kit is a complete setup" | It is missing the heater and the filter, which are the two items that decide the outcome                                                                                |
| "Flaring at a mirror is good exercise"          | It is a threat display, used deliberately in research to generate a measurable stress response                                                                          |
| "A betta only lives a year"                     | Two to four years is normal and five is reachable. The short lifespans are a husbandry result, not a species trait                                                      |

## Enrichment: what the research says

The betta is the most-studied fish in the freshwater hobby, and the findings are inconvenient for the way it is usually sold.

Restricted space raised signs of stress, including hiding and inactivity. Tank size and furnishings both measurably affected behavior and welfare. And fish in aquaria planted with *Sagittaria subulata* showed the longest average active swimming time and fewer negative behaviors.

The counterintuitive part is the planting. Adding dense cover does not produce a fish you never see. It produces one that swims more, because being out in open water is only a risk when there is nowhere else to be. Live planting also works on the water chemistry and the tank's microbial community at the same time, so it earns its place twice.

### Priority order

1. Tank volume, decided before anything else goes in it
2. Dense planting and cover, live where you can
3. Surface resting spots and broken sightlines
4. Varied feeding location and varied food
5. Tannins from driftwood, and softened light
6. Mirrors, at the bottom of the list or off it entirely

#### ON GROUP HOUSING

Research indicates bettas of either sex can be group-housed with lower aggression *if they have been reared in groups since hatching under enriched conditions*. That is a finding about rearing, not a license to combine adults. Two adult males in one tank will fight until one dies, whatever the tank is planted with.

#### DON'T READ A STILL FISH AS A CALM ONE

A betta sitting motionless in a bare tank is displaying the exact pattern the restricted-space study produced. Stillness in an empty box is not contentment, and it is the single easiest welfare problem in this species to miss.

#### TANNINS AND LIGHT

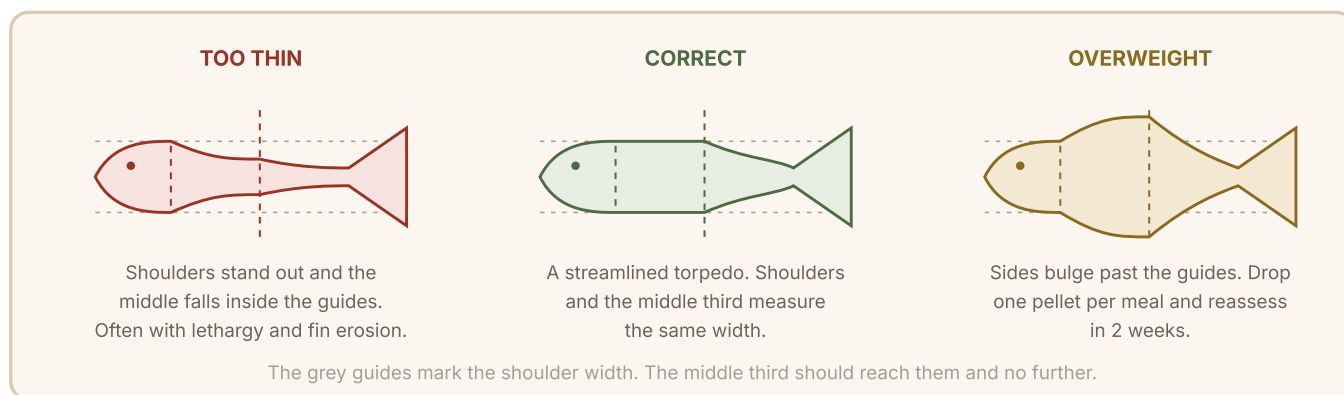
Driftwood releases tannins, which tint the water faintly amber and soften the light. That is closer to the blackwater conditions this species comes from, and it is cosmetic rather than harmful. If the tint bothers you, a shorter photoperiod or a dimmer light gets you most of the same effect. A betta lit hard from above in a bare tank has nowhere to be out of the glare.

# Reading a healthy betta

You will never handle this fish, so reading it from outside the glass is the whole skill.

## Body condition, from directly above

Look down into the tank from the top, not through the side. A betta in good condition looks like a streamlined torpedo, with the "shoulders" just behind the gill covers the same width as the middle third of the body.



## A healthy betta

- Deep, even color that holds day to day
- Fins carried open and flowing, edges intact
- Active swimming through the whole tank, not one corner
- Eager, immediate interest in food
- Regular trips to the surface for air, then back down
- Clear eyes, smooth scales, no lesions

## Stress signals

- Fins clamped tight against the body
- Faded color, or horizontal stress stripes
- Lethargy, hiding, or resting on the substrate
- Rapid gill movement, or gulping at the surface constantly
- Loss of appetite
- Scratching or flashing against decor

### NEVER BARE HANDS

A betta's slime coat is an active barrier against infection and parasites, and skin contact strips it. Move the fish with a net or, better, a cup, and only when you actually have to. When a new betta comes home, float the sealed bag for about 15 minutes to equalize temperature, then net or cup the fish across. Never pour store water into your tank.

## Health red flags & first response

Almost every illness in this section traces back to water quality or water temperature. That is genuinely good news, because it means the first move is nearly always the same.

### TEST THE WATER BEFORE YOU TREAT ANYTHING

Ammonia, nitrite, nitrate, and temperature, in that order, before a single dose of medication goes in the tank. A huge share of what looks like disease is a water-quality problem that will not respond to treatment, and medication on top of bad water is two problems instead of one. Correcting the water is not "doing nothing" while you wait. It is the treatment.

### Act today

| WHAT YOU SEE                                                  | WHAT IT POINTS AT                                                        |
|---------------------------------------------------------------|--------------------------------------------------------------------------|
| Any ammonia or nitrite reading above 0 ppm                    | Water emergency. Change 25 to 50% now and again tomorrow (page 9)        |
| Scales standing out from the body like a pinecone             | Dropsy, a sign of organ failure and generally a poor prognosis (page 20) |
| A fine gold or rust-colored dust over the body                | Velvet, which moves fast and kills fast (page 19)                        |
| White cottony or grey saddle-shaped patches, rapid spread     | Columnaris, highly contagious and fast (page 19)                         |
| Fins visibly shorter or more ragged than last week            | Fin rot, actively progressing (page 18)                                  |
| Grains of white salt across the body and fins                 | Ich (page 18)                                                            |
| Unable to stay upright, stuck floating or stuck on the bottom | Swim bladder disorder (page 20)                                          |
| Gasping at the surface constantly, gills working hard         | Ammonia or nitrite poisoning, or very low oxygen. Test immediately       |
| Open sores, ulcers, or a curved or S-shaped spine             | Possible mycobacteriosis (page 20). Use gloves from now on               |
| Refusing food 5 to 7 days with any other symptom present      | Investigate rather than wait (page 11)                                   |

## Finding a vet & what to tell them

Most betta problems are solved at the test kit. The ones that are not need somebody who has seen a fish before.

Fish veterinarians are far rarer than reptile or avian vets, and most betta problems genuinely are managed by correcting water quality rather than by treatment. That said, "rare" is not "nonexistent." Search for an aquatic veterinarian in your region now, while nothing is wrong, and write the number on page 24. Many will do a phone or video consult and advise on medication choice, which is often exactly what you need.

### What to have ready when you call

- Current ammonia, nitrite, nitrate, and pH, with the test kit type
- Current temperature, and whether it has been stable
- Tank volume, and how long it has been running
- Date of the last water change and how much you changed
- What the fish is doing, and when it changed
- What it is eating and how much, from the log on page 32
- Anything added recently: decor, plants, food, a tankmate
- Any medication already used, at what dose, and for how long

#### A HOSPITAL TANK IS WORTH HAVING

The veterinary reference build is a 10 gallon (38 L) tank with a sponge filter, an air pump and a heater. For one betta, a bare 2 to 3 gallon (8 to 11 L) container with a spare heater and a sponge filter seeded in your main tank does the same job. Either lets you treat a fish without dosing medication into your plants, your snails, and your bacteria colony. It is the cheapest piece of equipment on this list and the one most keepers wish they had bought before they needed it.

#### QUARANTINE EVERYTHING WET

New fish, plants, snails, and even decor from another tank can carry ich, velvet, and columnaris. Quarantine a new fish for at least 30 days in a separate container before it goes anywhere near an established tank, and 30 to 60 days for one you especially do not want to lose. Rinse and inspect plants. Never move a net, a siphon, or a bucket between tanks without drying it fully in between.

#### WHILE MEDICATING ANYTHING

Remove activated carbon from the filter, it adsorbs the medication out of the water. Keep the biological media in place, though. Do not combine medications unless a label or a vet says the combination is safe. Increase aeration, because most treatments reduce dissolved oxygen. And keep testing: medication can stall a bacteria colony and produce an ammonia spike on top of the illness.

## Fin rot & ich

### Fin rot

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>    | A bacterial infection of the fin tissue, almost always secondary to poor water quality, chronic cold, or a physical fin injury from sharp decor or a strong current                                                                                                                                                                                                                                                                        |
| <b>Signs</b>    | Fin edges frayed, ragged, or torn-looking. Black, brown, or milky discoloration along the margins. Fins visibly shorter week to week. In advanced cases the erosion reaches the body and the fin rays themselves                                                                                                                                                                                                                           |
| <b>Response</b> | Test and fix the water first, this is the actual cure in mild cases. Increase water change frequency. Remove anything sharp. Aquarium salt at about 1 tablespoon per 5 gallons (per 19 L), dissolved in a cup of tank water before it goes in, is the vet-reviewed starting dose, and it is for scaled fish only. Advanced cases, or any case not improving after about a week, need an antibacterial medication or veterinary antibiotics |
| <b>Recovery</b> | Regrowth begins as a clear or pale membrane at the fin edge, and takes weeks to months to fill back in. Regrown fin often comes back a different shade, sometimes clear, which is normal. If the edges stop receding, it is working                                                                                                                                                                                                        |

#### AQUARIUM SALT: THE LIMITS

There is no agreed number of days to stop at, so do not run a countdown. End it the way the veterinary guidance does: once the infection has cleared, change the water repeatedly until the salt is fully out. Salt does not evaporate, so between changes only replace what you remove, dosed to the volume replaced. It also harms most live plants, snails, and shrimp, which is what the hospital tank on page 17 is for. And aquarium salt is not table salt: no iodine, no anti-caking agents.

### Ich (white spot disease)

|                 |                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>    | The parasite <i>Ichthyophthirius multifiliis</i> , usually introduced on a new fish, plant, or unquarantined decor, and triggered into an outbreak by a chill or a stress event                                                                                                                                                                                               |
| <b>Signs</b>    | Small white grains, like scattered salt, across the body and fins. Flashing and scratching against decor, often days before the spots appear. Clamped fins, rapid gill movement, and appetite loss                                                                                                                                                                            |
| <b>Response</b> | Raise the temperature <i>gradually</i> , no more than 1 to 2°F (0.5 to 1°C) per hour, to around 82°F (28°C), which speeds the parasite through the only stage of its life cycle that medication can reach. Add a copper or formalin-based ich treatment and follow the full course. Warmer water holds less oxygen, so increase surface agitation while the temperature is up |
| <b>Recovery</b> | Spots fall away over roughly a week. Keep dosing for the full labeled course after the last visible spot, because the free-swimming stage is what you are actually killing. Then bring the temperature back down as slowly as you raised it                                                                                                                                   |

#### WHY THE TEMPERATURE TRICK WORKS

Ich is only vulnerable while it is swimming free between hosts. Encysted on the fish, it is shielded from medication; in its reproductive stage on the substrate it is shielded too. Warmth shortens the whole cycle so more of the population passes through that exposed window while your treatment is in the water.

## Velvet & columnaris

The two fastest killers on this list. Both are survivable if you catch them in the first day or two.

### Velvet (gold dust disease)

|                 |                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>    | <i>Piscinoodinium</i> , the freshwater velvet parasite and a dinoflagellate. <i>Oodinium</i> is its marine counterpart and the two names get swapped constantly. Usually introduced on a new fish and set off by stress, chilling, or poor water quality                                                                                                                                                 |
| <b>Signs</b>    | A fine gold, rust, or bronze dust over the body, easiest to see with a flashlight held at an angle in a darkened room. Clamped fins, flashing against decor, rapid gill movement, appetite loss, and lethargy. Often mistaken for ich, but the particles are far finer and the sheen is metallic rather than white                                                                                       |
| <b>Response</b> | Treat immediately, this one kills in days. Darken the tank completely: the parasite is partly photosynthetic and light feeds it. Dose a copper-based or antiparasitic velvet treatment for the full labeled course. Raise the temperature slightly if the label calls for it, and increase surface agitation. Remove carbon from any filter while medicating, it strips the medication straight back out |
| <b>Recovery</b> | The dust fades over several days. Keep the tank dark and finish the course after the last visible sign. Gills take the most damage, so expect laboured breathing to be the last symptom to resolve                                                                                                                                                                                                       |

### Columnaris

|                 |                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>    | <i>Flavobacterium columnare</i> , a bacterium that is highly contagious between fish and between tanks on shared equipment. It becomes far more aggressive in warm water, which is what makes it so dangerous in a tropical tank                                                                                                                                                                        |
| <b>Signs</b>    | White or greyish cottony patches, classically a saddle-shaped band across the back, or around the mouth, which is where the old name "mouth fungus" comes from. Frayed fin edges, ulcers, and pale gills. It is not a fungus, and it moves far faster than fin rot: hours to days rather than weeks                                                                                                     |
| <b>Response</b> | Lower the temperature toward the bottom of the betta's tolerable range, around 76°F (24.5°C), gradually. This is the opposite of the ich response and getting it backwards accelerates the infection. Isolate the fish. Treat with an antibacterial medication effective against gram-negative bacteria, or veterinary antibiotics. Disinfect nets, siphons, and buckets before they touch another tank |
| <b>Recovery</b> | Improvement should be visible within a couple of days of correct treatment. The aggressive strains can kill within 24 to 48 hours, so if there is no change on day two, escalate rather than waiting out the course                                                                                                                                                                                     |

#### ICH OR VELVET OR COLUMNARIS?

Distinct white grains, like salt, evenly scattered: ich. A fine metallic gold or rust sheen best seen under a torch: velvet. A white or grey cottony patch, often saddled across the back or around the mouth, spreading visibly by the hour: columnaris. When you are unsure between ich and velvet, treat as velvet, since it is the faster of the two and the treatments overlap. Never guess between ich and columnaris, because the temperature instructions point in opposite directions.

# Swim bladder, dropsy & mycobacteriosis

## Swim bladder disorder

|                 |                                                                                                                                                                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>    | Usually overfeeding or constipation, a swollen gut pressing on the swim bladder. Sometimes infection, injury, or a defect in heavily bred strains                                                                                                                                            |
| <b>Signs</b>    | Buoyancy problems: floating at the surface, sinking to the bottom, listing to one side, or swimming at an odd angle. Often with a visibly bloated belly and stringy or absent feces                                                                                                          |
| <b>Response</b> | Fast for 2 to 3 days. Confirm the temperature is in range, since cold slows digestion. Then offer one small piece of blanched, shelled pea, or daphnia, to move things along. Lower the water level slightly if the fish struggles to reach the surface, and add a resting spot near the top |
| <b>Recovery</b> | Mild constipation cases resolve in a few days. When they do, go back to page 10 and cut the portion permanently, because it will recur otherwise. Cases that persist beyond a week, or come with other symptoms, are more likely infection or a structural problem and need a vet            |

## Dropsy

|                 |                                                                                                                                                                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>    | Not a disease in itself. It is the visible result of fluid building up inside the body from organ failure, most often the kidneys, usually secondary to long-term infection or chronic poor conditions                                                                 |
| <b>Signs</b>    | A swollen belly with the scales standing out from the body, giving the classic pinecone look when viewed from above. Lethargy, appetite loss, and often bulging eyes                                                                                                   |
| <b>Response</b> | Isolate, keep the water pristine and the temperature stable, and speak to an aquatic vet. By the time the pinecone sign appears the underlying failure is usually advanced and the prognosis poor. Humane euthanasia advised by a vet is a legitimate and kind outcome |
| <b>Recovery</b> | Uncommon. What dropsy is genuinely useful for is as a warning about the tank: it is a long-term-conditions disease, so review pages 6 to 9 carefully before adding another fish                                                                                        |

## Mycobacteriosis (fish tuberculosis)

|                 |                                                                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>    | <i>Mycobacterium</i> species, present at low levels in many aquariums and becoming a problem in stressed or immunocompromised fish. Rare, but worth knowing because it is zoonotic                                                      |
| <b>Signs</b>    | Slow decline: wasting despite eating, a curved or S-shaped spine, open sores, faded color, lost scales. Weeks to months rather than days                                                                                                |
| <b>Response</b> | There is no reliable cure in home aquaria. Isolate the fish, keep the equipment for that tank strictly separate, and talk to a vet about prognosis and humane euthanasia. Do not move fish or equipment from that tank into another one |
| <b>Recovery</b> | Rare. Manage this one as a containment problem rather than a treatment problem                                                                                                                                                          |

### ZOONOTIC: PROTECT YOURSELF

*Mycobacterium marinum* transmits to people through aquarium water, especially through broken skin, causing a stubborn infection sometimes called fish tank granuloma. Wear waterproof gloves if you have cuts or a suspected case in the tank, wash your hands after every session regardless, and never siphon by mouth. If a persistent lesion appears on a hand or forearm, tell your doctor you keep an aquarium: it changes what they test for.

## Stress signals & behavior

The behavior problems in this species are mostly housing problems, and they show up on the fish before they show up in the water.

### Reading the fish

| BEHAVIOR                                | WHAT IT USUALLY MEANS                                                                                        | FIRST THING TO CHECK                           |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Clamped fins held tight to the body     | General stress, and one of the earliest visible signs of a water problem                                     | Test the water, then the temperature           |
| Faded color, or dark horizontal stripes | Stress response. Horizontal stress stripes are distinct from the vertical breeding bars on females           | Water, temperature, and anything new nearby    |
| Resting on the substrate or on a leaf   | Normal in an older fish or one with heavy finnage, especially near a hammock. Concerning if new and constant | Whether there is a surface resting spot at all |
| Gulping at the surface repeatedly       | Normal, that is the labyrinth organ. Constant, urgent gulping is not                                         | Ammonia and nitrite, and surface agitation     |
| Flaring at its own reflection           | A threat display, not exercise. Tank glass in a bright room can reflect all day                              | Room lighting, and what is behind the tank     |
| Hiding constantly in a bare tank        | The restricted-space pattern from the research on page 14                                                    | Tank volume, planting, and cover               |
| Glass surfing, pacing the front pane    | Stress, or a reflection it is responding to                                                                  | Reflections, flow, and tank size               |
| Building a bubble nest                  | A comfortable, well-warmed male. Nothing to correct                                                          | Nothing. This is the good one                  |

### Flaring, and what to do about it

Flaring is the full threat display: gill covers spread, fins fanned, body turned side-on. It is normal behavior and a fish that never flares is not necessarily happier. What matters is what it is aimed at, and how often.

- A few seconds at a passing shadow, at you, or at a new decor item is ordinary and passes.
- Repeated flaring at one spot on the glass is a reflection. Move the light, put something behind the tank, or reduce the contrast between a bright tank and a dark room.
- Flaring that runs for long stretches every day costs the fish energy and appetite. Treat it as a stressor to remove, not a display to enjoy.
- A mirror is not enrichment. If you want to see the display once, a few seconds is enough, and it belongs at the bottom of the list on page 14.

#### THE ORDER TO CHECK THINGS IN

Whatever the behavior, the sequence is the same: water first, temperature second, then what changed in or around the tank in the last week, then the fish itself. Working in that order stops you treating a fish for an illness it does not have.

## Tankmates

A betta does not need company and is not lonely alone. Anything you add is for you.

Anything you add is a risk you are choosing to take on the fish's behalf, so add it deliberately or not at all. The absolute rules are short: one male per tank, never two, and never a male with a female outside a supervised, deliberate breeding attempt.

| OPTION                                       | VERDICT                                                                                                                                                                 |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alone, in a planted 5 to 10 gal (19 to 38 L) | <b>RECOMMENDED</b> The default, and the setup this whole package is written around                                                                                      |
| Nerite snail or mystery snail                | <b>USUALLY FINE</b> In 5 gal and up. Some bettas pick at snail antennae, so watch the first week                                                                        |
| Amano or ghost shrimp                        | <b>RISKY</b> 10 gal and up with heavy cover. Some bettas treat shrimp as food and there is no undoing that                                                              |
| Small schooling fish, corydoras              | <b>RISKY</b> 15 to 20 gal (57 to 76 L) minimum, never fin-nippers, and only with a genuine plan for separating them                                                     |
| Another male betta                           | <b>NEVER</b> They will fight until one dies. There is no tank size that solves this                                                                                     |
| A "sorority" of females                      | <b>NOT FOR BEGINNERS</b> Frequently ends in serious injury. The group-housing research is about fish reared together from hatching, not adults introduced to each other |
| Guppies or other long-finned fish            | <b>NEVER</b> A betta reads flowing fins as a rival, and guppies get shredded                                                                                            |

### IF YOU DO ADD ANYTHING

Quarantine it for at least 30 days first (page 17), add it to the betta's tank rather than the other way around only if you must, rearrange the decor first so the territory feels new to both, and have a second cycled container genuinely ready before the introduction, not planned for afterward. Watch continuously for the first hour and then daily for a week.

## Signs an introduction is failing

- The betta shadows or follows the newcomer continuously rather than losing interest after the first hour.
- Nipped fins, missing scales, or a snail withdrawn into its shell and staying there.
- Either animal permanently occupying one corner, or the betta unable to reach the surface without passing a confrontation.
- The betta stops eating, or its color fades, in the days after the addition.

Any of these means separating them, not waiting to see. Damage in a small tank happens over hours, and a betta that has decided something is a rival does not usually change its mind.

## Setup checklist & water targets

### Essential setup checklist

- ☐ 5 gal (19 L) tank minimum, 10 gal (38 L) better
- ☐ Tight-fitting lid, with the air gap left above the water
- ☐ Adjustable submersible heater, 15 to 25 W
- ☐ Separate thermometer, never just the heater dial
- ☐ Sponge filter, air pump, airline and a check valve
- ☐ Liquid reagent test kit, not paper strips
- ☐ Dechlorinator that handles chloramine too
- ☐ Sand or smooth gravel, about 1 in (2.5 cm)
- ☐ Live or silk plants, driftwood, and a cave
- ☐ A surface resting spot: leaf hammock or broad-leaved plant
- ☐ Gravel vacuum and two aquarium-only buckets
- ☐ Betta pellet, and a net or cup for transfers

### Water targets

| PARAMETER    | TARGET                                                                                   | ACT IF                                 |
|--------------|------------------------------------------------------------------------------------------|----------------------------------------|
| Temperature  | 78 to 80°F (25.5 to 26.5°C)                                                              | Outside 76 to 82°F (24.5 to 28°C)      |
| Ammonia      | 0 ppm                                                                                    | Anything above 0                       |
| Nitrite      | 0 ppm                                                                                    | Anything above 0                       |
| Nitrate      | Under 20 ppm                                                                             | Over 20 ppm                            |
| pH           | 6.5 to 7.5                                                                               | Swinging, more than sitting off-target |
| Water change | 20 to 30% weekly, temperature-matched and dechlorinated, with a partial substrate vacuum |                                        |

### Before your betta comes home

- ☐ Tank fully cycled: 2 ppm ammonia gone to 0 ammonia and 0 nitrite in 24 hours
- ☐ Nitrate brought under 20 ppm with a large water change
- ☐ Heater running 24 to 48 hours at a stable, verified 78 to 80°F (25.5 to 26.5°C)
- ☐ Aquatic vet or consult service found, and the number written on page 24

## Emergency & quick targets

### ACT TODAY IF

Ammonia or nitrite above 0 ppm · scales standing out like a pinecone · a fine gold or rust dust over the body · white or grey cottony patches spreading by the hour · fins shorter than last week · white grains across body and fins · stuck floating or on the bottom · constant urgent gasping at the surface · open sores or a curved spine · five to seven days of refused food with any other symptom

| READING      | TARGET                                                   |
|--------------|----------------------------------------------------------|
| Temperature  | 78 to 80°F (25.5 to 26.5°C)                              |
| Ammonia      | 0 ppm                                                    |
| Nitrite      | 0 ppm                                                    |
| Nitrate      | Under 20 ppm                                             |
| pH           | 6.5 to 7.5, stable                                       |
| Feeding      | 2 to 4 pellets, once or twice daily, one fast day a week |
| Water change | 20 to 30% weekly, dechlorinated, temperature-matched     |

### Your numbers

Aquatic vet or consult \_\_\_\_\_

Tank volume and heater wattage \_\_\_\_\_

Local fish store that knows me \_\_\_\_\_

Date the tank finished cycling \_\_\_\_\_

Fish's name and arrival date \_\_\_\_\_

Last water change \_\_\_\_\_

### HAVE THIS READY WHEN YOU CALL

Ammonia, nitrite, nitrate, pH, and the kit used. Current temperature and whether it has been stable. Tank volume and how long it has run. Last water change, date and volume. What the fish is doing and when it changed. Anything added recently, and any medication given, at what dose and for how long.

### THE ONE MOVE THAT FIXES MOST THINGS

A 25 to 50% temperature-matched, dechlorinated water change. It is the right first response to almost every emergency on this page, and it buys time while you work out what is happening.

## SECTION 05 · OWNER TOOLS

# Budget & shopping list

US retail ranges. The low end is basic but correct gear, the high end is premium.

| INITIAL SETUP                                 | TYPICAL COST          |
|-----------------------------------------------|-----------------------|
| 5 to 10 gallon tank with a lid                | \$30 to \$90          |
| Adjustable submersible heater, 15 to 25 W     | \$15 to \$30          |
| Sponge filter, air pump, airline, check valve | \$18 to \$40          |
| Thermometer, glass or digital                 | \$5 to \$12           |
| Substrate: sand or smooth gravel              | \$10 to \$20          |
| Live or silk plants, driftwood, and a cave    | \$20 to \$40          |
| Leaf hammock or surface resting spot          | \$4 to \$10           |
| Liquid reagent test kit                       | \$22 to \$35          |
| Dechlorinator and bottled nitrifying bacteria | \$12 to \$22          |
| Gravel vacuum and two aquarium-only buckets   | \$15 to \$25          |
| Betta pellet, small container                 | \$6 to \$12           |
| <b>Equipment subtotal</b>                     | <b>\$157 to \$336</b> |
| The betta itself                              | \$5 to \$30           |
| <b>All-in before the fish comes home</b>      | <b>\$162 to \$366</b> |

Rare varieties reach about \$55, and males cost more than females for finnage, not care. A hospital container (page 17) adds \$25 to \$45.

| MONTHLY ONGOING                                   | TYPICAL COST        |
|---------------------------------------------------|---------------------|
| Food, averaged over the year                      | \$2 to \$4          |
| Dechlorinator                                     | \$1 to \$3          |
| Filter sponge and air stone replacement, averaged | \$1 to \$2          |
| Test kit refills, averaged                        | \$2 to \$4          |
| Electricity for the heater and air pump           | \$3 to \$8          |
| Medication reserve, averaged                      | \$1 to \$4          |
| <b>Monthly ongoing</b>                            | <b>\$10 to \$25</b> |

Roughly \$120 to \$300 a year. Food is almost never the expensive part: a betta eats a few pellets a day. Heating is, at \$30 to \$100 a year depending on the climate and the room.

### WHERE NOT TO CUT CORNERS

The heater and the test kit. The heater is the parameter that makes every other parameter more forgiving, and the test kit is the only reason you will know a problem exists before the fish shows it. Save on decor and substrate before you save on either. The bowl-and-fish bundle is missing exactly these two, which is why it fails.

## First 30 days checklist

Day 1 here is the day the fish arrives, which is weeks after the day the tank was set up.

### Before day 1

- ☐ Tank cycled and confirmed: 2 ppm ammonia to 0 ammonia and 0 nitrite in 24 hours (page 7)
- ☐ Heater verified at 78 to 80°F (25.5 to 26.5°C) with your own thermometer for 24 to 48 hours

### Day 1

- ☐ Float the sealed bag for about 15 minutes, then net or cup the fish across. Never pour the store water in
- ☐ Lights off, no feeding, and leave the fish alone for the rest of the day
- ☐ Photograph the fish from the side and from above, as a baseline for color, fins, and body condition
- ☐ Fill in the numbers on page 24 and the equipment dates on page 33

### Week 1

- ☐ Offer 2 to 3 pellets on day 2. A refusal in the first few days is ordinary and is not a symptom yet
- ☐ Test ammonia and nitrite every 1 to 2 days: a new fish is a new bioload and the colony has to catch up
- ☐ Watch for flashing, clamped fins, white spots, or a gold sheen, the signs a shop-tank problem came home too
- ☐ No decor changes, no rearranging, no tankmates

### Week 2 to 3

- ☐ Settle into the feeding routine from page 10, including the weekly fasting day
- ☐ First full weekly water change, 20 to 30%, temperature-matched (page 9)
- ☐ Start the Owner Log on page 32 and check body condition from above against page 15
- ☐ Move the feeding spot around the tank rather than using the same corner

### Week 4

- ☐ Drop testing to weekly once ammonia and nitrite have read 0 for a full week
- ☐ Review the log for drift in temperature, appetite, or color rather than judging any single day
- ☐ Add planting or cover from page 34 now that the fish has settled, one change at a time

### Days 31 to 90

- ☐ Compare a fresh top-down photo against your day 1 one and adjust the portion if the shape has changed
- ☐ Replace test reagents before they expire, and check the heater still holds its set point
- ☐ Fill in the pet-sitter sheet on page 31 before you need it, and set up the blackout kit on page 29
- ☐ Start the monthly enrichment rotation on page 34

## Symptom quick reference

| SYMPTOM                                         | LIKELY CAUSE                           | ACTION                                                                      |
|-------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------|
| White grains like salt across body and fins     | Ich                                    | Raise slowly to 82°F (28°C), copper or formalin course (page 18)            |
| Fine gold or rust dust, seen under a torch      | Velvet                                 | Darken the tank, copper treatment, act today (page 19)                      |
| White or grey cottony saddle, spreading fast    | Columnaris                             | Lower toward 76°F (24.5°C), isolate, antibacterial (page 19)                |
| Fins frayed, ragged, or shorter than last week  | Fin rot                                | Fix the water, salt at 1 tbsp per 5 gal, cleared by water changes (page 18) |
| Scales standing out like a pinecone             | Dropsy, organ failure                  | Isolate, pristine water, vet. Prognosis poor (page 20)                      |
| Floating, sinking, or listing to one side       | Swim bladder disorder                  | Fast 2 to 3 days, then a pea. Cut portions after (page 20)                  |
| Bloated belly, stringy or absent feces          | Overfeeding or constipation            | Fast, check temperature, pre-soak pellets (pages 10 and 20)                 |
| Wasting despite eating, curved spine, sores     | Possible mycobacteriosis               | Isolate, gloves from now on, vet (page 20)                                  |
| Constant urgent gasping at the surface          | Ammonia, nitrite, or low oxygen        | Test now, 25 to 50% water change, add agitation (page 8)                    |
| Flashing or scratching against decor            | Parasites, or an irritant in the water | Test, then look closely for ich or velvet (pages 18 and 19)                 |
| Clamped fins, faded color, lethargy             | Stress, usually water or temperature   | Test the water and the temperature before anything else (page 16)           |
| Refusing food, otherwise normal                 | New tank, cold water, or picky         | Work the six-reason list, then wait (page 11)                               |
| Refusing food 5 to 7 days with a second symptom | Several possible causes                | Investigate now, bring the Owner Log (page 32)                              |
| Hiding constantly in a bare tank                | Restricted space and no cover          | Add volume, planting and cover (page 14)                                    |
| Glass surfing or flaring all day                | Reflections, flow, or too little space | Change the lighting or backdrop, check flow (page 21)                       |
| Torn fin with a clean edge                      | A physical tear, not an infection      | Remove the sharp decor, keep the water pristine (page 13)                   |

## Daily, weekly & seasonal routine

### Daily

- ☐ Read the thermometer before you do anything else
- ☐ Feed the measured portion, moving the spot around, and remove anything uneaten
- ☐ Thirty seconds of watching: fins, color, breathing, and how much of the tank it is using
- ☐ Check the filter is still bubbling and the surface is moving

### Weekly

- ☐ Test ammonia, nitrite, and nitrate, on the same day each week, and log it on page 32
- ☐ Change 20 to 30%, dechlorinated and temperature-matched, vacuuming a different section of substrate
- ☐ One fasting day, and a top-down look at body condition (page 15)
- ☐ Wipe the glass and trim or replace any dying plant matter

### Monthly

- ☐ Rinse the filter sponge in old tank water only if flow has dropped. Never under the tap
- ☐ Verify the heater still holds its set point against your own thermometer
- ☐ Review the log for slow trends rather than single readings
- ☐ Rotate one piece of enrichment from page 34

### Every 3 to 6 months

- ☐ Replace the air stone and check the airline and check valve for cracks
- ☐ Check test kit reagent expiry dates. Expired reagents read wrong, which is worse than not testing

### Seasonal

- ☐ Winter: the room cools, so check the heater is keeping up and not just running longer
- ☐ Summer: watch for the tank running above 82°F (28°C) in a hot room, and add surface agitation if it does

### Yearly

- ☐ Replace the heater on schedule rather than on failure. They fail on, which is the dangerous way
- ☐ Replace the filter sponge, keeping the old one in the tank alongside it for a few weeks first

## Power outages & the blackout plan

Oxygen goes first and temperature second, even for a fish that can breathe air. Aerate before you do anything else.

### When the power goes out

#### Do

- **Aerate straight away.** Once the filter stops, the surface stops moving and dissolved oxygen starts falling within a couple of hours. A battery or USB air pump on a power bank, left running, is the whole job.
- No pump: scoop water out and pour it back from a few inches up, every 10 to 15 minutes at this tank size. Crude, and it works.
- The labyrinth organ is your margin, not your plan. A betta takes oxygen at the surface, so it tolerates still water far better than a gill-only fish. Keep aerating anyway.
- Insulate. Wrap the tank in towels or blankets, keep the lid on, and leave the front uncovered so you can still see the fish.
- Heat gently: a sealed hot-water bottle or a hand warmer taped to the *outside* of the glass, checking the thermometer as you go.
- Unplug the filter at the wall so it cannot flush stagnant, rotting media into the tank when power returns.
- Stop feeding entirely, and keep the tank dark and quiet.

#### Don't

- Don't feed. Food raises oxygen demand and adds ammonia to water with no filter running.
- Don't put a heat source in the water or in direct contact with the fish.
- Don't do a water change during the outage, and don't restart a filter that has sat unpowered for hours until you have rinsed the media in tank water.
- Don't let the recovery be abrupt: bring temperature back gradually, then test ammonia and nitrite before you feed again.

#### BLACKOUT KIT

A battery or USB air pump with a spare air stone, a power bank, hand warmers, two thick towels, a jug, and the numbers from page 24.

### Rough timeline

| ELAPSED       | WHAT TO DO                                                                                                                                               |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| First 10 min  | Get air moving: battery pump on, or start the pour-back cycle. Unplug the filter at the wall, lid on, lights off                                         |
| 0 to 2 hours  | Keep aerating. No feeding. Temperature has barely moved yet, so do not chase it                                                                          |
| 2 to 8 hours  | Insulate with towels and start watching the thermometer. Aeration continues throughout                                                                   |
| Past 24 hours | Still aerating, still no feeding. If the tank cannot be held above about 72°F (22°C), move the fish somewhere that can                                   |
| Power back    | Rinse the filter media in tank water before restarting it. Bring temperature back gradually, test ammonia and nitrite, then wait 24 hours before feeding |

## Travel, transport & moving

A betta travels better than most fish, because it can breathe the air in the bag.

### Leaving town

| TRIP LENGTH   | WHAT TO DO                                                                                                                                                                                                                                          |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 to 3 nights | Fine alone. A healthy adult skips meals routinely and one fasting day is already part of the schedule. Water change the day before you leave, check the heater, and go. Do not use a holiday feeder block, which fouls water badly in a small tank. |
| 4 to 7 nights | A sitter every 2 to 3 days to feed the pre-portioned meals, read the thermometer, and look at the fish. Or an automatic feeder loaded with exact portions, plus one sitter visit mid-trip.                                                          |
| Over a week   | A sitter every 2 to 3 days who has done a practice visit with you, including one water change if the trip runs past a week. Leave page 31 filled in and the meals pre-portioned.                                                                    |

### Transport and moving

- Short trips, under about 6 hours: a bag or a sealed container, one third to halfway full with the fish's own tank water, the rest air. Bettas breathe from the surface, so the air gap is the point.
- Longer trips: a lidded bucket with an air stone on a battery pump holds temperature and oxygen far better than a bag. A shop can also fill a transport bag with pure oxygen, which extends the safe window considerably.
- Insulate the container in a cooler or a padded box, keep it out of direct sun, and do not run the car heater or air conditioning straight onto it. Do not feed for 24 hours before travel.
- Moving house: transport as much of the original tank water and the filter sponge as you can, in separate sealed containers, keeping the sponge submerged the whole time. That sponge is your cycle. Set the tank up and get it warm before the fish goes back in, and test daily for the first week.

### The first week after a move

- ☐ Test ammonia and nitrite daily. Moving disturbs the substrate and the media, and a mini-cycle is common
- ☐ Skip the first day's feeding, then feed lightly for several days
- ☐ Check the heater is holding: a new room is a new ambient temperature
- ☐ Expect faded color, hiding, and a refused meal or two. Give it a week before treating it as a symptom
- ☐ Leave the decor arranged the way it was, if you can. Familiar layout helps a fish settle
- ☐ Keep the tank out of direct sun and away from a draft or a radiator in the new spot

#### NEVER

Move a betta in a sealed bag with no air space, transport it in water it has not been living in, or leave the container in a parked car. And never set the tank up with entirely fresh water because the old water "looked dirty." The filter sponge carries the cycle, but the tank water buys you a gentler transition.

## FILL IN AND LEAVE BESIDE THE TANK

# Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person before the first solo visit.

Fish's name \_\_\_\_\_

Aquatic vet or fish store \_\_\_\_\_

Tank volume and where the food is \_\_\_\_\_

Backup contact \_\_\_\_\_

My phone \_\_\_\_\_

Days to feed on \_\_\_\_\_

## Each visit

- ☐ Read the thermometer. It should say 78 to 80°F (25.5 to 26.5°C). Write it down.
- ☐ Check the filter is still bubbling and the water surface is moving.
- ☐ Find the fish. It should be swimming, with its fins open. Coming up to the surface for air is normal and healthy.
- ☐ Feed only on the marked days, and only the pre-portioned amount left out. This is a very small amount of food and that is correct.
- ☐ If anything is floating uneaten after five minutes, scoop it out with the net.
- ☐ Wash your hands after, and don't rinse anything in the kitchen sink.

## Please don't

- Feed extra, or feed on an unmarked day. Overfeeding is the fastest way to harm this fish, and it will always act hungry.
- Add tap water, change the water, or top the tank up unless I have written instructions for it. Untreated tap water is harmful here.
- Change any dial or setting on the heater, the pump, or the light timer.
- Tap the glass, hold a mirror up, or put anything into the tank.
- Take the lid off and leave it off. Bettas jump.

### CALL ME IF

The thermometer reads below 74°F (23°C) or above 84°F (29°C) · the filter has stopped bubbling · the fish is lying on the bottom, stuck at the surface, or swimming on its side · its fins are clamped tight to its body or look ragged · it has white spots, a gold dusty sheen, or white fuzzy patches · its belly is swollen or its scales stick out · the water has turned cloudy or smells bad · it has not eaten on a feeding day

## Owner log

You cannot weigh a betta at home, so the test results and your own written description are the only trend data you will ever have. A nitrate reading creeping up week on week tells you the water change is too small before the fish tells you. Sixteen rows is four months, which is long enough to see a real trend rather than a bad day.

# Equipment, water change & vet log

## Equipment log

| ITEM                                              | INSTALLED | REPLACE OR CHECK BY | DONE |
|---------------------------------------------------|-----------|---------------------|------|
| Heater (verify set point monthly, replace yearly) |           |                     |      |
| Air pump, airline and check valve                 |           |                     |      |
| Filter sponge (overlap the new one with the old)  |           |                     |      |
| Air stone, every 3 to 6 months                    |           |                     |      |
| Test kit reagents, check the expiry dates         |           |                     |      |

## Twelve-month planner

Tick the box in the month you did it. Start with whichever month you're in.

| TASK                        | M1                       | M2                       | M3                       | M4                       | M5                       | M6                       | M7                       | M8                       | M9                       | M10                      | M11                      | M12                      |
|-----------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Verify heater set point     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Review the log for trends   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Air stone and airline check | <input type="checkbox"/> |                          |                          | <input type="checkbox"/> |                          |                          | <input type="checkbox"/> |                          |                          | <input type="checkbox"/> |                          |                          |
| Reagent expiry check        | <input type="checkbox"/> |                          |                          |                          |                          | <input type="checkbox"/> |                          |                          |                          |                          | <input type="checkbox"/> |                          |
| Replace heater              | <input type="checkbox"/> |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          | <input type="checkbox"/> |

## Quarantine log (any new fish, plant, or snail, 30 days minimum)

| DATE IN | DATE OUT | WHAT IT WAS, AND ANYTHING NOTICED |
|---------|----------|-----------------------------------|
|         |          |                                   |

## Treatment & vet log

| DATE | SYMPTOM OR DIAGNOSIS | TREATMENT, DOSE AND DURATION | OUTCOME |
|------|----------------------|------------------------------|---------|
|      |                      |                              |         |
|      |                      |                              |         |

## Enrichment checklist & log

### Setup

- ☐ Dense planting, live where possible, soft silk where not
- ☐ A surface resting spot: leaf hammock or a broad-leaved plant near the top
- ☐ Driftwood or decor positioned to break the tank into zones and cut sightlines
- ☐ At least one cave or fully enclosed hide with a smooth entrance

### Monthly review

- ☐ Move or add one item, without blocking the route to the surface
- ☐ Rotate the food and the feeding spot, and try one new safe item from page 12
- ☐ Check the fish is using more of the tank than last month, not less
- ☐ Run a hand over every decor item. If it would snag a stocking, it will tear a fin

### Enrichment log

| DATE | WHAT CHANGED | HOW IT WENT, AND WHETHER TO REPEAT |
|------|--------------|------------------------------------|
|      |              |                                    |
|      |              |                                    |
|      |              |                                    |
|      |              |                                    |
|      |              |                                    |
|      |              |                                    |
|      |              |                                    |
|      |              |                                    |
|      |              |                                    |

#### WHAT NOT TO DO

Don't use a mirror as a daily routine: what you are provoking is a threat display, and it is what researchers use to generate a measurable stress response. Don't use stiff plastic plants. Don't add so much decor that the fish cannot reach the surface, because air access outranks everything on this page. And don't change three things at once, or you will never know which one the fish responded to.

## REFERENCE

# Glossary

|                                 |                                                                                                                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ammonia (NH<sub>3</sub>)</b> | The first and most toxic product of fish waste and decaying food. Target is 0 ppm, always.                                                                                                     |
| <b>Bubble nest</b>              | A raft of saliva-coated air bubbles a male builds at the surface to hold eggs. Captive males build them with no female present, and it generally signals a comfortable fish.                   |
| <b>Chloramine</b>               | Chlorine bonded to ammonia, used by many water utilities. It does not gas off the way chlorine does, so the dechlorinator has to be one that handles it.                                       |
| <b>Clamped fins</b>             | Fins held tight against the body instead of spread. One of the earliest visible signs of stress or a water problem.                                                                            |
| <b>Cycled</b>                   | A tank whose bacteria colonies convert a 2 ppm ammonia dose to 0 ammonia and 0 nitrite within 24 hours. The single prerequisite for adding a fish.                                             |
| <b>Dropsy</b>                   | Not a disease but a sign: fluid retention from organ failure, showing as a swollen body with scales standing out like a pinecone.                                                              |
| <b>Flashing</b>                 | Scratching or rubbing against decor and substrate. Usually the first sign of a parasite, often days before anything is visible.                                                                |
| <b>GH and KH</b>                | General hardness (dissolved minerals) and carbonate hardness (buffering capacity). Secondary parameters, but very low KH lets pH crash after a water change.                                   |
| <b>Hospital tank</b>            | A bare container with a spare heater and a seeded sponge, used to treat a fish without dosing medication into plants, snails, and the main filter colony.                                      |
| <b>Labyrinth organ</b>          | A folded, vascularized structure above the gills that lets a betta breathe air from the surface. It is why surface access is a physiological requirement.                                      |
| <b>Nitrite (NO<sub>2</sub>)</b> | The middle step of the nitrogen cycle. Blocks oxygen uptake in the blood. Target is 0 ppm.                                                                                                     |
| <b>Nitrate (NO<sub>3</sub>)</b> | The end product of a healthy cycle. Mildly toxic and removed by water changes. Keep under 20 ppm.                                                                                              |
| <b>Ovipositor (egg spot)</b>    | A small white spot between the ventral and anal fins on a female, the tube eggs pass through. A male can show a protruding vent that resembles it, so it is suggestive rather than conclusive. |
| <b>ppm</b>                      | Parts per million, the unit every test kit on page 8 reports in. Equivalent to milligrams per liter.                                                                                           |
| <b>Seeded media</b>             | Filter sponge or gravel taken from an established, healthy tank, carrying a live bacteria colony. The fastest legitimate way to shorten a cycle.                                               |
| <b>Slime coat</b>               | The mucus layer over a fish's body, an active barrier against infection and parasites. Bare hands strip it, which is why bettas are moved by net or cup.                                       |
| <b>Sponge filter</b>            | An air-driven filter whose sponge is the surface the bacteria live on. The standard for bettas because it filters without producing current.                                                   |
| <b>Stress stripes</b>           | Dark horizontal bands appearing along a stressed betta's body. Distinct from the vertical bars a receptive female shows.                                                                       |
| <b>Swim bladder</b>             | The gas-filled organ controlling buoyancy. When compressed by a swollen gut, the fish floats, sinks, or lists.                                                                                 |

## REFERENCE

# Sources

Where the numbers in this package come from, so you can check them yourself.

## Veterinary and scientific sources

- Life in a fishbowl: space and environmental enrichment affect behaviour of *Betta splendens*. *Animal Welfare*.
- Life beyond a jar: effects of tank size and furnishings on the behaviour and welfare of Siamese fighting fish.
- Effects of aquatic plants on water quality, microbial community and fish behaviors in newly established Betta aquaria.
- Timing of isolation from an enriched environment determines aggressive behavior and sexual maturity in *Betta splendens*. *BMC Zoology*, 2021.
- Merck Veterinary Manual, Aquarium Fish, for disease presentation and the zoonotic risk on page 20.
- Merck Veterinary Manual, Management of Aquarium Fish and Routine Health Care of Fish, for the 30 day quarantine minimum and the 30 to 60 day range, the 10 gallon hospital tank build, and the carbon and aeration rules while medicating, on pages 17 and 22.
- Merck Veterinary Manual, Parasitic Diseases of Fish, for *Piscinoodinium* as the freshwater velvet genus on page 19.
- VCA Animal Hospitals, Fish White Spot Disease (Ich), for the aquarium salt dose of about a tablespoon per 5 gallons, for scaled fish only, and for ending a salt course by water change rather than on a fixed schedule (page 18).
- PetMD, Betta Fish Care Sheet. Reviewed veterinary guidance on husbandry, feeding, and common conditions.
- California Fish Veterinary Service, on assessing betta body condition from above and correcting it by portion (page 15).

## Husbandry references

- Aquarium Co-Op, betta fish care guide and its aquarium power outage guidance, for the filter-media handling on page 29.
- NC State College of Veterinary Medicine, on aquariums and ponds during power outages, for aeration coming ahead of temperature and the manual agitation interval on page 29.
- Practical Fishkeeping, frequently asked questions on fishless cycling and its complete cycling guide, for the cycle timings, the completion test, and the fish-in action line on page 7.
- Bettafish.org, food and feeding, for portioning and the treat schedule on pages 10 and 12.
- Beastly Facts betta fish guides: tank setup, water parameters, feeding, health issues, handling, cost, and enrichment, current at the time of publication.

REFERENCE

# Version history & about

What changed in each edition, and the small print.

## Version history

| VERSION        | TEMPLATE | WHAT CHANGED                                                                                                                                                                                                                                                                                                                                               |
|----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.2 · Sep 2026 | t3       | Cross-checked against the new site articles on aquarium cycling, power outages and transport, and fish quarantine and treatment. Quarantine raised to the Merck 30 day minimum, the 10 day salt ceiling replaced with the veterinary endpoint rule, the blackout plan reordered to put aeration first, and the fish-in action line set at 0.1 to 0.25 ppm. |
| 2.1 · Sep 2026 | t3       | The shared t3 spacing rules the other guides on this template use were applied to this file, where they had been missing. No words and no type sizes changed, and no content differs from 2.0.                                                                                                                                                             |
| 2.0 · Sep 2026 | t3       | Corrected against fresh sourcing. Feeding window moved from about 60 seconds to 1 to 2 minutes, which is what the vet-reviewed guidance actually says. Running costs corrected upward to \$10 to \$25 a month and \$120 to \$300 a year after the heater electricity line was found to be understated.                                                     |
| 1.0 · Sep 2026 | t3       | First edition. 36 pages on the t3 package template, adapted to the aquatic layout: housing collapsed into tank, filtration and cycling, and water quality, with a full fishless cycling walkthrough, a top-down body condition diagram, and both units throughout.                                                                                         |

This package is a husbandry reference compiled from the sources above and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for an aquatic veterinarian, and nothing in it should delay professional advice. Free companion guides, updated as guidance changes, are at [beastlyfacts.com](#).

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